MATH60633A – TP2: Options Risk Management

Team

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Contents

1 0. Data Loading & Exploration **1**

2 1. Pricing the Portfolio of Options [5 pts] **3**

3 2. One Risk Driver — Univariate Gaussian Model [10 pts] **5**

4 3. Two Risk Drivers — Bivariate Gaussian Model [15 pts] **7**

5 4. Two Risk Drivers — Copula-Marginal Model [20 pts] **8**

6 5. Volatility Surface [20 pts] **10**

6.1 5.1 Fit the Parametric Surface 10

6.2 5.2 Re-price Portfolio Using the Volatility Surface 12

7 6. Summary & Comparison **14**

1 0. Data Loading & Exploration

```
library(xts)
library(MASS)

source("R/functions.R")
load("Market.rda")

# Extract key objects
S0      <- as.numeric(tail(Market$sp500, 1))
vix0    <- as.numeric(tail(Market$vix, 1))
rf_mats <- as.numeric(names(Market$rf))
rf_vals <- as.numeric(Market$rf)

cat("Latest S&P 500 price:", S0, "\n")
```

```
## Latest S&P 500 price: 1683.99
```

```r
cat("Latest VIX (decimal):", vix0, "(", vix0 * 100, "%)\n")
```

```
## Latest VIX (decimal): 0.1453 ( 14.53 %)
```

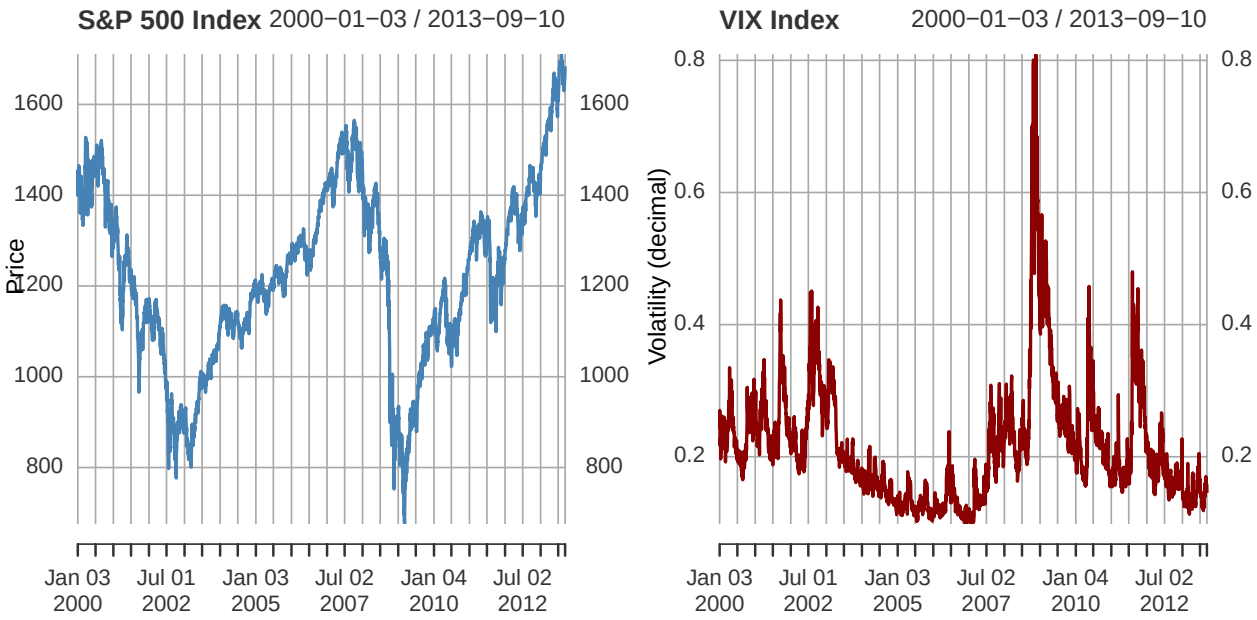
```r
par(mfrow = c(2, 2))

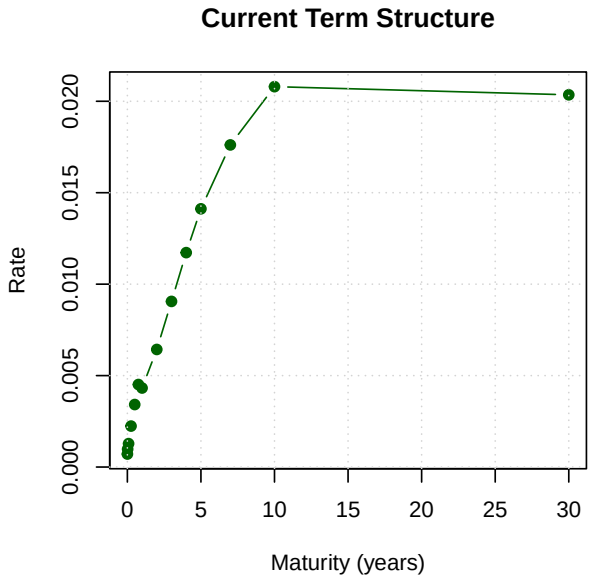
# S&P 500 time series
plot(Market$sp500, main = "S&P 500 Index", ylab = "Price", col = "steelblue")

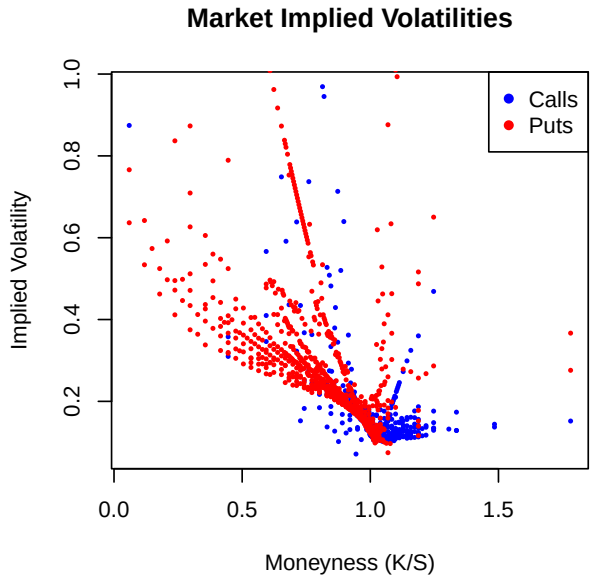
# VIX time series
plot(Market$vix, main = "VIX Index", ylab = "Volatility (decimal)", col = "darkred")

# Term structure
plot(rf_mats, rf_vals, type = "b", pch = 19, col = "darkgreen",
     xlab = "Maturity (years)", ylab = "Rate",
     main = "Current Term Structure")
grid()

# Options implied volatility vs moneyness
m_calls <- Market$calls[, "K"] / S0
m_puts  <- Market$puts[, "K"] / S0
plot(m_calls, Market$calls[, "IV"], pch = 16, cex = 0.5, col = "blue",
     xlab = "Moneyness (K/S)", ylab = "Implied Volatility",
     main = "Market Implied Volatilities", xlim = range(c(m_calls, m_puts)))
points(m_puts, Market$puts[, "IV"], pch = 16, cex = 0.5, col = "red")
legend("topright", legend = c("Calls", "Puts"), col = c("blue", "red"), pch = 16)
```







2 1. Pricing the Portfolio of Options [5 pts]

We define a book of four European call options:

Option	Strike (K)	Maturity (days)
1	1600	20
2	1650	20
3	1750	40
4	1800	40

Conventions: 360 days/year for the term structure, 250 days/year for option maturity.

```r
# Interpolated rates
r_20d <- interpolate_rate(rf_mats, rf_vals, 20 / 360)
r_40d <- interpolate_rate(rf_mats, rf_vals, 40 / 360)
cat("Interpolated rate (20-day):", r_20d, "\n")
```

```
## Interpolated rate (20-day): 0.001149957
```

```r
cat("Interpolated rate (40-day):", r_40d, "\n")
```

```
## Interpolated rate (40-day): 0.001439839
```

```r
# Option time-to-maturity
tau1 <- 20 / 250
tau2 <- 40 / 250
cat("tau_opt (20-day):", tau1, "years\n")
```

```
## tau_opt (20-day): 0.08 years
```

```r
cat("tau_opt (40-day):", tau2, "years\n")
```

```
## tau_opt (40-day): 0.16 years
```

```r
# Price each option using Black-Scholes with VIX as implied vol
c1 <- bs_call(S0, 1600, r_20d, vix0, tau1)
c2 <- bs_call(S0, 1650, r_20d, vix0, tau1)
c3 <- bs_call(S0, 1750, r_40d, vix0, tau2)
c4 <- bs_call(S0, 1800, r_40d, vix0, tau2)

# Portfolio value
V0 <- c1 + c2 + c3 + c4

cat("\n--- Individual Option Prices ---\n")
```

```
##
## --- Individual Option Prices ---
```

```r
cat("C(K=1600, T=20d):", round(c1, 4), "\n")
```

```
## C(K=1600, T=20d): 87.5694
```

```r
cat("C(K=1650, T=20d):", round(c2, 4), "\n")
```

```
## C(K=1650, T=20d): 47.7236
```

```r
cat("C(K=1750, T=40d):", round(c3, 4), "\n")
```

```
## C(K=1750, T=40d): 15.3022
```

```r
cat("C(K=1800, T=40d):", round(c4, 4), "\n")
```

```
## C(K=1800, T=40d): 6.3798
```

```r
cat("\nPortfolio value V0:", round(V0, 4), "\n")
```

```
##
## Portfolio value V0: 156.975
```

3 2. One Risk Driver — Univariate Gaussian Model [10 pts]

We model the daily log-returns of the S&P 500 as iid normally distributed. We simulate 10,000 scenarios for the one-week ahead (5 business days) underlying price.

```r
# Daily log-returns of S&P 500
sp_logret <- diff(log(as.numeric(Market$sp500)))

mu_sp    <- mean(sp_logret)
sigma_sp <- sd(sp_logret)

cat("Daily log-return mean:", mu_sp, "\n")
```

```
## Daily log-return mean: 4.283042e-05
```

```r
cat("Daily log-return std: ", sigma_sp, "\n")
```

```
## Daily log-return std:  0.01332592
```

```r
cat("Number of observations:", length(sp_logret), "\n")
```

```
## Number of observations: 3409
```

```r
n_sim  <- 10000
n_days <- 5

# Simulate 5 daily log-returns per scenario, then sum
Z_sp <- matrix(rnorm(n_sim * n_days, mean = mu_sp, sd = sigma_sp),
               nrow = n_sim, ncol = n_days)
cum_logret_sp <- rowSums(Z_sp)

# Project S&P price one week ahead
S_T5 <- S0 * exp(cum_logret_sp)
```

```r
# Re-price portfolio at T+5: maturities decrease by 5 days
# VIX stays constant (single risk driver model)
V_T5 <- portfolio_value(S_T5, vix0, rf_mats, rf_vals, days1 = 15, days2 = 35)

PnL_uni <- V_T5 - V0
```

```r
# Risk measures
VaR95_uni <- -quantile(PnL_uni, 0.05)
ES95_uni  <- -mean(PnL_uni[PnL_uni <= -VaR95_uni])

cat("--- Univariate Gaussian Risk Measures ---\n")
```

```
## --- Univariate Gaussian Risk Measures ---
```

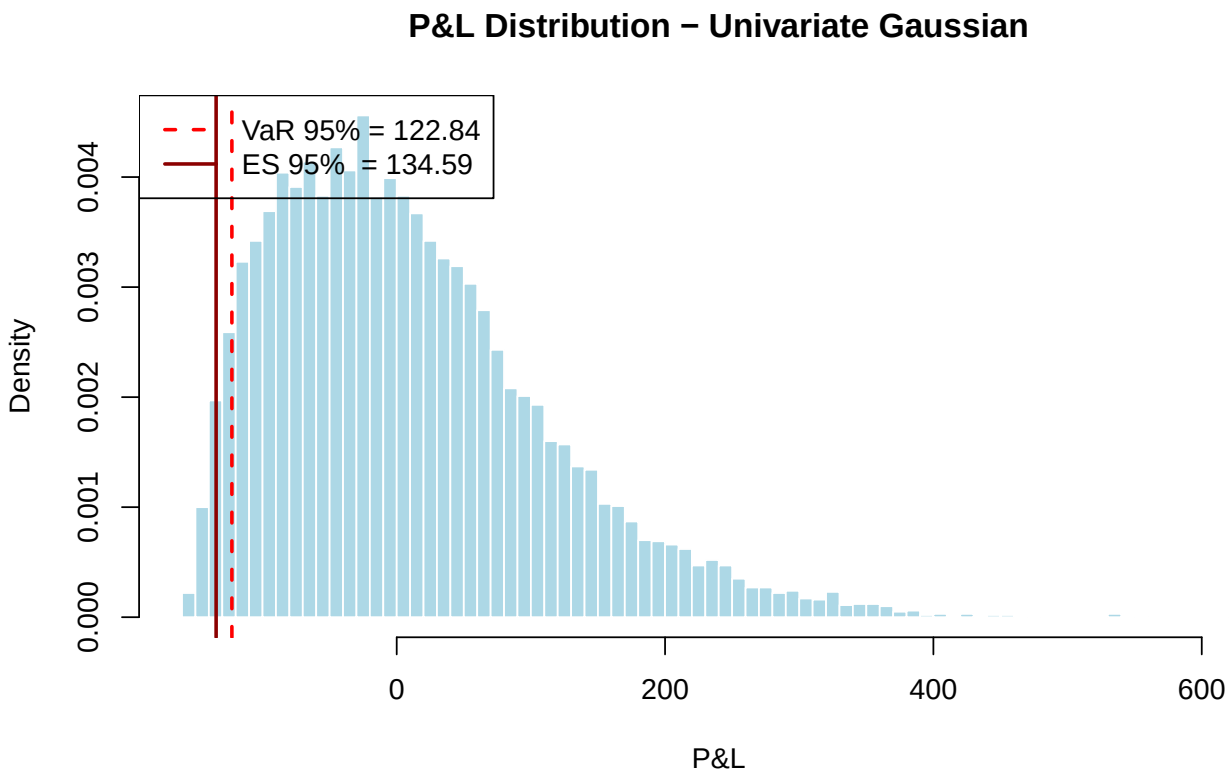
```r
cat("VaR 95%:", round(VaR95_uni, 4), "\n")
```

```
## VaR 95%: 122.8448
```

```r
cat("ES  95%:", round(ES95_uni, 4), "\n")
```

```
## ES  95%: 134.5857
```

```r
hist(PnL_uni, breaks = 80, col = "lightblue", border = "white",
     main = "P&L Distribution - Univariate Gaussian",
     xlab = "P&L", probability = TRUE)
abline(v = -VaR95_uni, col = "red", lwd = 2, lty = 2)
abline(v = -ES95_uni, col = "darkred", lwd = 2, lty = 1)
legend("topleft", legend = c(paste("VaR 95% =", round(VaR95_uni, 2)),
                             paste("ES 95%  =", round(ES95_uni, 2))),
       col = c("red", "darkred"), lwd = 2, lty = c(2, 1))
```



4 3. Two Risk Drivers — Bivariate Gaussian Model [15 pts]

We add the VIX as a second risk driver. Both daily log-returns are modeled as jointly normally distributed.

```
# Daily log-returns of VIX
vix_logret <- diff(log(as.numeric(Market$vix)))

mu_vix    <- mean(vix_logret)
sigma_vix <- sd(vix_logret)
rho_biv   <- cor(sp_logret, vix_logret)

cat("VIX log-return mean:", mu_vix, "\n")
```

```
## VIX log-return mean: -0.0001497654
```

```
cat("VIX log-return std: ", sigma_vix, "\n")
```

```
## VIX log-return std:  0.0636733
```

```
cat("Correlation (S&P, VIX):", rho_biv, "\n")
```

```
## Correlation (S&P, VIX): -0.753651
```

```
mu_vec    <- c(mu_sp, mu_vix)
Sigma_mat <- matrix(c(sigma_sp^2,                      rho_biv * sigma_sp * sigma_vix,
                      rho_biv * sigma_sp * sigma_vix,  sigma_vix^2),
                    nrow = 2)

# Simulate 5 days of bivariate normal draws
Z_biv <- array(0, dim = c(n_sim, n_days, 2))
for (d in 1:n_days) {
  Z_biv[, d, ] <- mvrnorm(n_sim, mu = mu_vec, Sigma = Sigma_mat)
}

# Cumulative 5-day log-returns
cum_sp_biv  <- rowSums(Z_biv[, , 1])
cum_vix_biv <- rowSums(Z_biv[, , 2])

# Project prices
S_T5_biv   <- S0   * exp(cum_sp_biv)
VIX_T5_biv <- vix0 * exp(cum_vix_biv)
```

```
# Price portfolio at T+5 with simulated VIX as implied vol
V_T5_biv <- portfolio_value(S_T5_biv, VIX_T5_biv, rf_mats, rf_vals,
                            days1 = 15, days2 = 35)
PnL_biv  <- V_T5_biv - V0

VaR95_biv <- -quantile(PnL_biv, 0.05)
ES95_biv  <- -mean(PnL_biv[PnL_biv <= -VaR95_biv])

cat("--- Bivariate Gaussian Risk Measures ---\n")
```

```
## --- Bivariate Gaussian Risk Measures ---
```

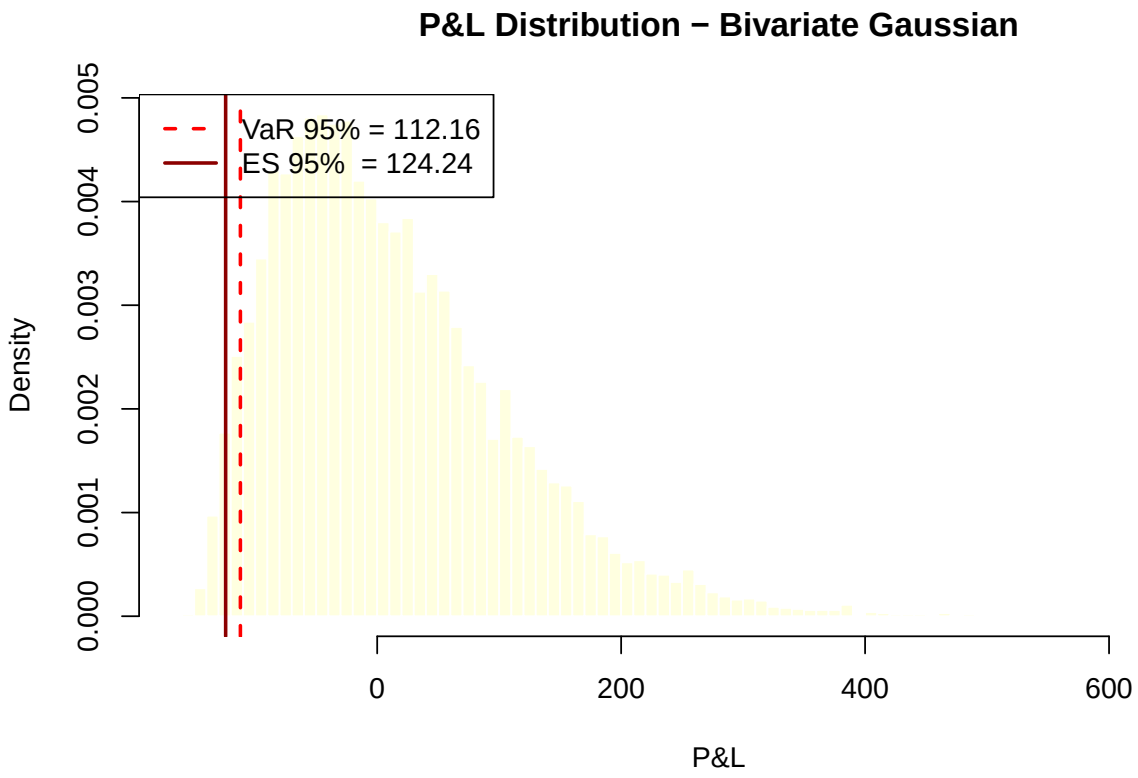
```
cat("VaR 95%:", round(VaR95_biv, 4), "\n")
```

```
## VaR 95%: 112.1629
```

```
cat("ES  95%:", round(ES95_biv, 4), "\n")
```

```
## ES  95%: 124.2383
```

```
hist(PnL_biv, breaks = 80, col = "lightyellow", border = "white",
     main = "P&L Distribution - Bivariate Gaussian",
     xlab = "P&L", probability = TRUE)
abline(v = -VaR95_biv, col = "red", lwd = 2, lty = 2)
abline(v = -ES95_biv, col = "darkred", lwd = 2, lty = 1)
legend("topleft", legend = c(paste("VaR 95% =", round(VaR95_biv, 2)),
                             paste("ES 95%  =", round(ES95_biv, 2))),
       col = c("red", "darkred"), lwd = 2, lty = c(2, 1))
```



5 4. Two Risk Drivers — Copula-Marginal Model [20 pts]

We use Student-t marginals ($\nu = 10$ for S&P, $\nu = 5$ for VIX) with a Gaussian copula to capture the dependence structure while allowing heavier tails.

```r
# Estimate Gaussian copula correlation via Kendall's tau inversion
rho_cop <- fit_gauss_copula(sp_logret, vix_logret)
cat("Kendall's tau:", cor(sp_logret, vix_logret, method = "kendall"), "\n")
```

```
## Kendall's tau: -0.5882059
```

```r
cat("Fitted Gaussian copula correlation:", rho_cop, "\n")
```

```
## Fitted Gaussian copula correlation: -0.7979894
```

```r
df_sp  <- 10
df_vix <- 5

# Standard deviations of Student-t distributions
sd_t_sp  <- sqrt(df_sp / (df_sp - 2))    # sqrt(10/8)
sd_t_vix <- sqrt(df_vix / (df_vix - 2))  # sqrt(5/3)

# Simulate 5 days of copula-based draws
Z_cop <- array(0, dim = c(n_sim, n_days, 2))
for (d in 1:n_days) {
  U_sim <- rgauss_copula(n_sim, rho_cop)

  # Transform to Student-t marginals
  z1_raw <- qt(U_sim[, 1], df = df_sp)
  z2_raw <- qt(U_sim[, 2], df = df_vix)

  # Scale to match data mean and standard deviation
  z1_scaled <- mu_sp  + (sigma_sp  / sd_t_sp)  * z1_raw
  z2_scaled <- mu_vix + (sigma_vix / sd_t_vix) * z2_raw

  Z_cop[, d, ] <- cbind(z1_scaled, z2_scaled)
}

# Cumulative 5-day log-returns
cum_sp_cop  <- rowSums(Z_cop[, , 1])
cum_vix_cop <- rowSums(Z_cop[, , 2])

# Project prices
S_T5_cop   <- S0   * exp(cum_sp_cop)
VIX_T5_cop <- vix0 * exp(cum_vix_cop)
```

```r
V_T5_cop <- portfolio_value(S_T5_cop, VIX_T5_cop, rf_mats, rf_vals,
                            days1 = 15, days2 = 35)
PnL_cop  <- V_T5_cop - V0

VaR95_cop <- -quantile(PnL_cop, 0.05)
ES95_cop  <- -mean(PnL_cop[PnL_cop <= -VaR95_cop])

cat("--- Copula-Marginal Risk Measures ---\n")
```

```
## --- Copula-Marginal Risk Measures ---
```

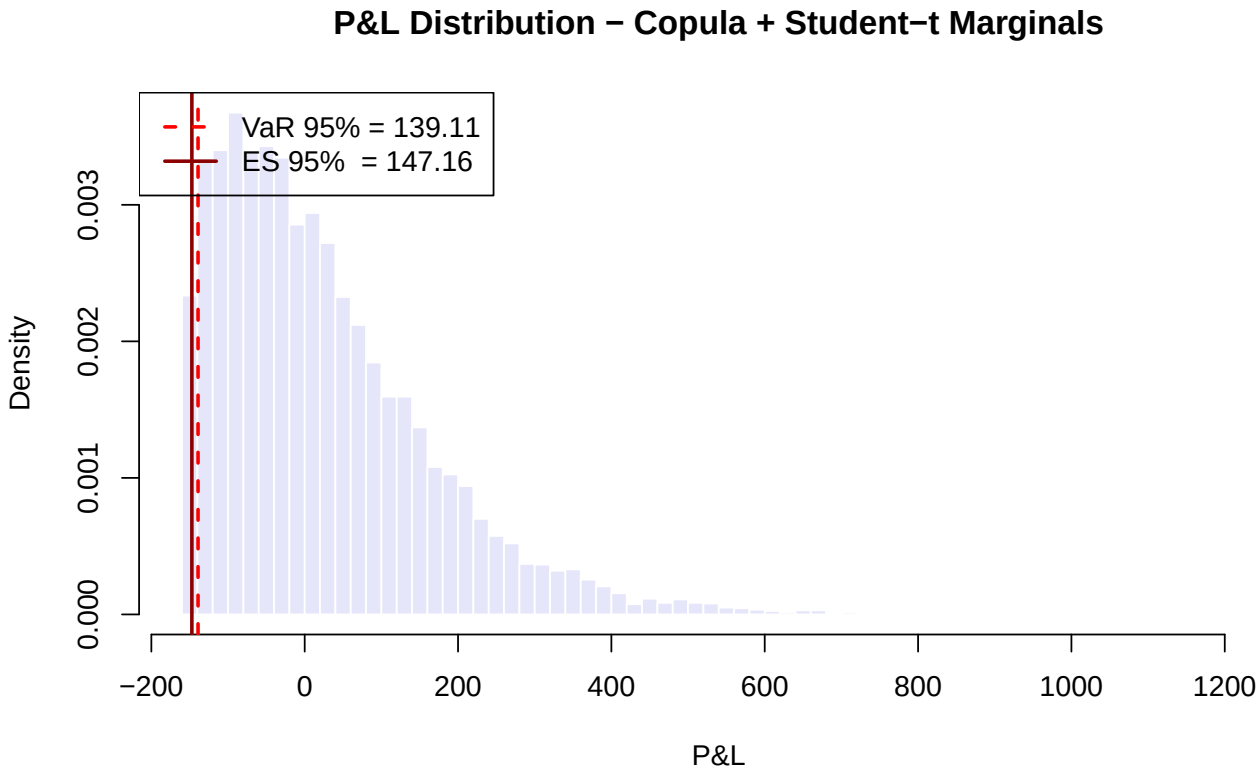
```r
cat("VaR 95%:", round(VaR95_cop, 4), "\n")
```

```
## VaR 95%: 139.1106
```

```r
cat("ES  95%:", round(ES95_cop, 4), "\n")
```

```
## ES  95%: 147.164
```

```r
hist(PnL_cop, breaks = 80, col = "lavender", border = "white",
     main = "P&L Distribution - Copula + Student-t Marginals",
     xlab = "P&L", probability = TRUE)
abline(v = -VaR95_cop, col = "red", lwd = 2, lty = 2)
abline(v = -ES95_cop, col = "darkred", lwd = 2, lty = 1)
legend("topleft", legend = c(paste("VaR 95% =", round(VaR95_cop, 2)),
                             paste("ES 95%  =", round(ES95_cop, 2))),
       col = c("red", "darkred"), lwd = 2, lty = c(2, 1))
```



6 5. Volatility Surface [20 pts]

6.1 5.1 Fit the Parametric Surface

We fit the parametric volatility surface:

$$\sigma(m, \tau) = \alpha_1 + \alpha_2(m-1)^2 + \alpha_3(m-1)^3 + \alpha_4\sqrt{\tau}$$

where $m = K/S$ is moneyness and τ is time to maturity in years. We minimize the sum of absolute deviations between market and model implied volatilities.

```r
# Combine calls and puts
all_opts <- rbind(Market$calls, Market$puts)
K_all    <- all_opts[, "K"]
tau_all  <- all_opts[, "tau"]
IV_all   <- all_opts[, "IV"]
m_all    <- K_all / S0

cat("Total options for surface fitting:", nrow(all_opts), "\n")
```

```
## Total options for surface fitting: 1172
```

```r
cat("Unique maturities:", length(unique(tau_all)), "\n")
```

```
## Unique maturities: 11
```

```r
cat("Moneyness range:", round(range(m_all), 4), "\n")
```

```
## Moneyness range: 0.0594 1.7815
```

```r
# L1 objective function
obj_fn <- function(alpha) {
  sigma_model <- alpha[1] + alpha[2] * (m_all - 1)^2 +
                 alpha[3] * (m_all - 1)^3 + alpha[4] * sqrt(tau_all)
  sum(abs(IV_all - sigma_model))
}

# Optimize with Nelder-Mead
fit_surface <- optim(par = c(0.15, 0.5, -0.1, 0.05), fn = obj_fn,
                     method = "Nelder-Mead",
                     control = list(maxit = 50000))

alpha <- fit_surface$par
cat("\nFitted parameters:\n")
```

```
##
## Fitted parameters:
```

```r
cat("  alpha1:", alpha[1], "\n")
```

```
##   alpha1: 0.2064946
```

```r
cat("  alpha2:", alpha[2], "\n")
```

```
##   alpha2: 1.560182
```

```r
cat("  alpha3:", alpha[3], "\n")
```

```
##   alpha3: 1.097558
```

```r
cat("  alpha4:", alpha[4], "\n")
```

```
##   alpha4: -0.05446395
```

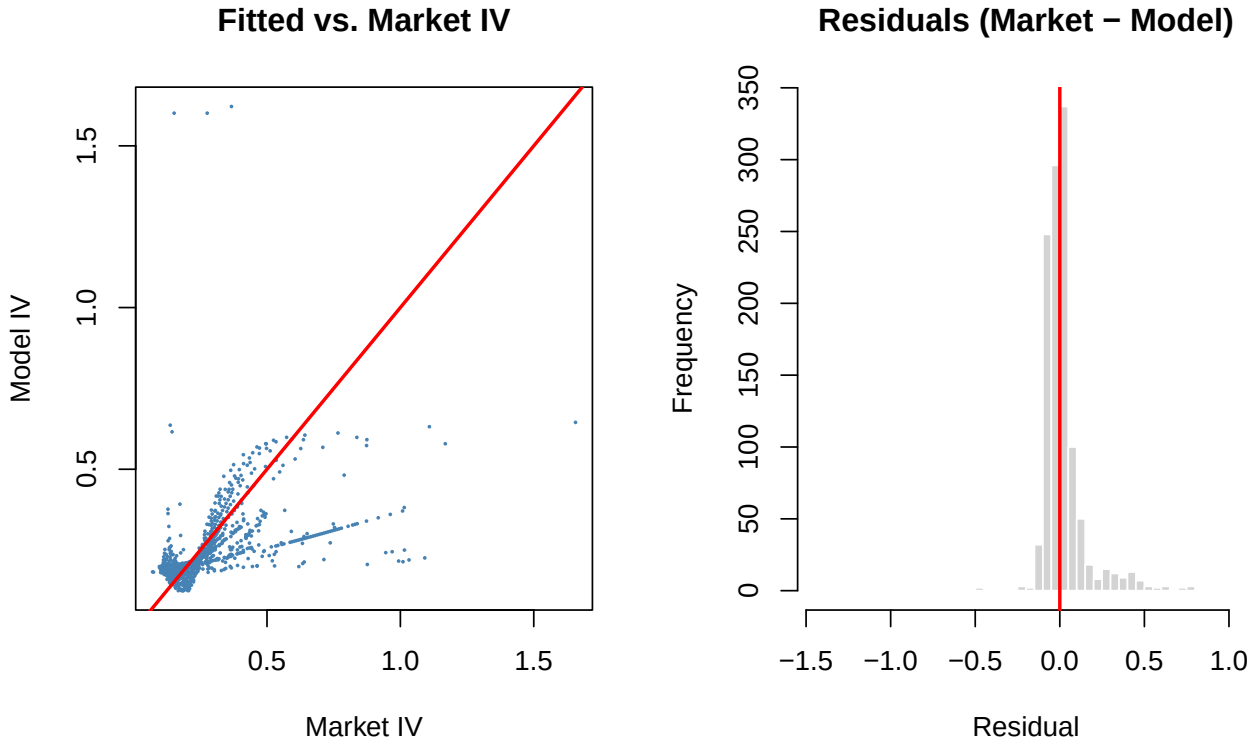
```r
cat("  Convergence:", fit_surface$convergence, "\n")
```

```
##   Convergence: 0
```

```r
# Plot fitted vs. market implied vols
IV_fitted <- sigma_hat(m_all, tau_all, alpha)
resid     <- IV_all - IV_fitted

par(mfrow = c(1, 2))
plot(IV_all, IV_fitted, pch = 16, cex = 0.3, col = "steelblue",
     xlab = "Market IV", ylab = "Model IV",
     main = "Fitted vs. Market IV")
abline(0, 1, col = "red", lwd = 2)

hist(resid, breaks = 60, col = "lightgray", border = "white",
     main = "Residuals (Market - Model)", xlab = "Residual")
abline(v = 0, col = "red", lwd = 2)
```



6.2 5.2 Re-price Portfolio Using the Volatility Surface

At time T, we use:

$$\tilde{\sigma}(m, \tau) = \sigma(m, \tau) - (\alpha_1 + \alpha_4) + \text{VIX}_T$$

At time $T+5$, we use:

$$\tilde{\sigma}(m,\tau) = \sigma(m,\tau) - (\alpha_1 + \alpha_4) + \text{VIX}_{T+5}$$

where VIX_{T+5} is obtained by simulation (copula model from Section 4).

```
# Portfolio value at T using volatility surface
V0_surf <- portfolio_value_surface(S0, vix0, rf_mats, rf_vals,
                                   days1 = 20, days2 = 40, alpha = alpha)
cat("Portfolio value at T (surface):", round(V0_surf, 4), "\n")
```

```
## Portfolio value at T (surface): 182.9509
```

```
# Portfolio value at T+5 using simulated S and VIX from Section 4
V_T5_surf <- portfolio_value_surface(S_T5_cop, VIX_T5_cop, rf_mats, rf_vals,
                                     days1 = 15, days2 = 35, alpha = alpha)

PnL_surf  <- V_T5_surf - V0_surf

VaR95_surf <- -quantile(PnL_surf, 0.05)
ES95_surf  <- -mean(PnL_surf[PnL_surf <= -VaR95_surf])

cat("\n--- Volatility Surface Risk Measures ---\n")
```

```
##
## --- Volatility Surface Risk Measures ---
```

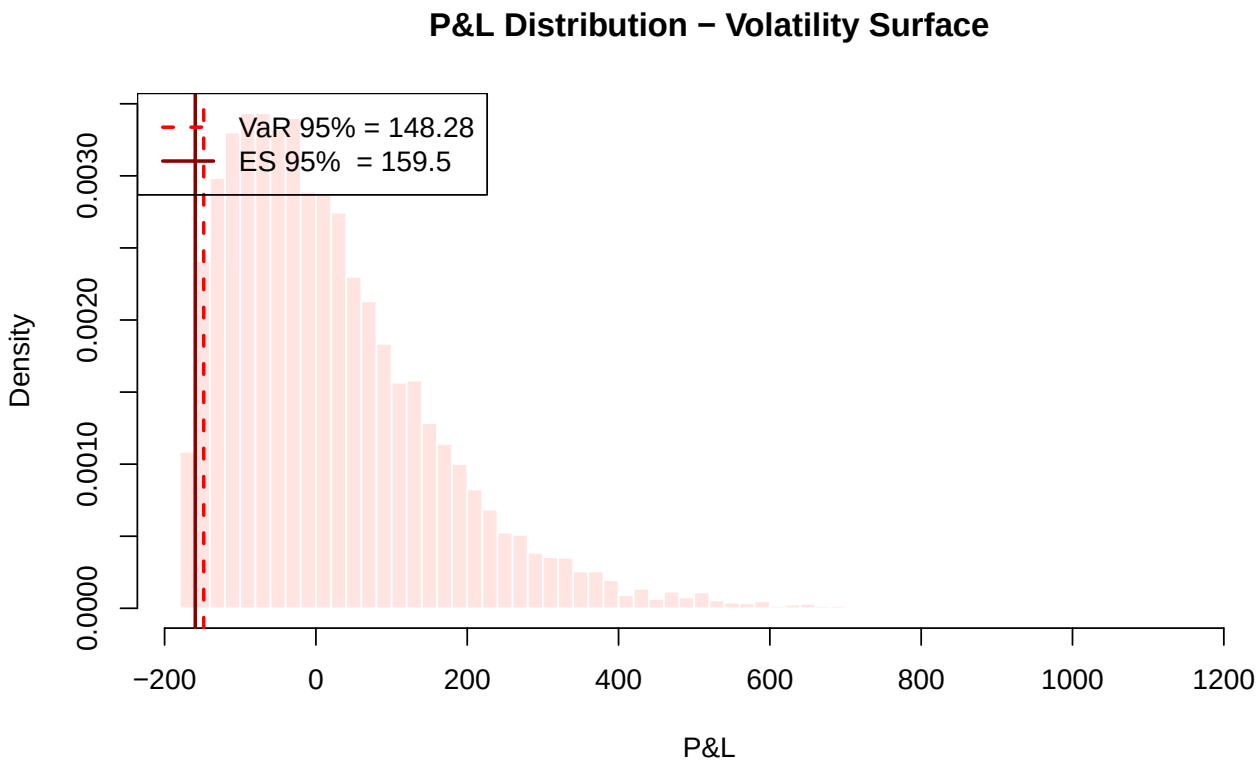
```
cat("VaR 95%:", round(VaR95_surf, 4), "\n")
```

```
## VaR 95%: 148.2783
```

```
cat("ES  95%:", round(ES95_surf, 4), "\n")
```

```
## ES  95%: 159.496
```

```
hist(PnL_surf, breaks = 80, col = "mistyrose", border = "white",
     main = "P&L Distribution - Volatility Surface",
     xlab = "P&L", probability = TRUE)
abline(v = -VaR95_surf, col = "red", lwd = 2, lty = 2)
abline(v = -ES95_surf, col = "darkred", lwd = 2, lty = 1)
legend("topleft", legend = c(paste("VaR 95% =", round(VaR95_surf, 2)),
                             paste("ES 95%  =", round(ES95_surf, 2))),
       col = c("red", "darkred"), lwd = 2, lty = c(2, 1))
```



7 6. Summary & Comparison

```
results <- data.frame(
  Model   = c("Univariate Gaussian", "Bivariate Gaussian",
              "Copula + Student-t", "Volatility Surface"),
  VaR_95  = round(c(VaR95_uni, VaR95_biv, VaR95_cop, VaR95_surf), 2),
  ES_95   = round(c(ES95_uni, ES95_biv, ES95_cop, ES95_surf), 2)
)
knitr::kable(results, col.names = c("Model", "VaR 95%", "ES 95%"),
             caption = "Risk Measures Comparison")
```

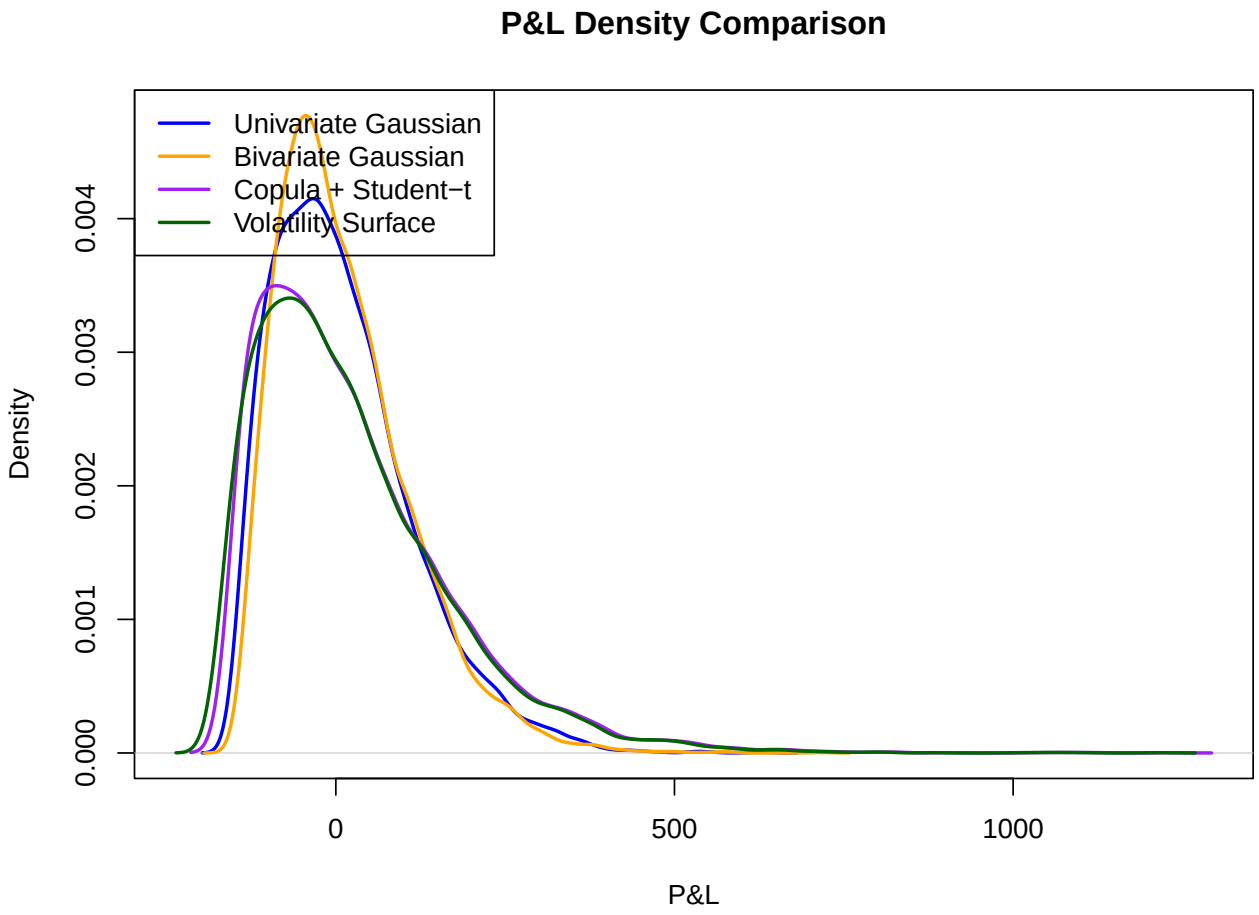
Table 1: Risk Measures Comparison

Model	VaR 95%	ES 95%
Univariate Gaussian	122.84	134.59
Bivariate Gaussian	112.16	124.24
Copula + Student-t	139.11	147.16
Volatility Surface	148.28	159.50

```
# Overlay P&L densities
d_uni  <- density(PnL_uni)
d_biv  <- density(PnL_biv)
d_cop  <- density(PnL_cop)
d_surf <- density(PnL_surf)
```

```r
xlim <- range(c(d_uni$x, d_biv$x, d_cop$x, d_surf$x))
ylim <- range(c(d_uni$y, d_biv$y, d_cop$y, d_surf$y))

plot(d_uni, col = "blue", lwd = 2, xlim = xlim, ylim = ylim,
     main = "P&L Density Comparison", xlab = "P&L")
lines(d_biv, col = "orange", lwd = 2)
lines(d_cop, col = "purple", lwd = 2)
lines(d_surf, col = "darkgreen", lwd = 2)
legend("topleft",
       legend = c("Univariate Gaussian", "Bivariate Gaussian",
                  "Copula + Student-t", "Volatility Surface"),
       col = c("blue", "orange", "purple", "darkgreen"),
       lwd = 2)
```



Key observations:

1. The **univariate Gaussian** model only captures delta risk (S&P movements). It ignores volatility changes, leading to potentially understated tail risk.
2. The **bivariate Gaussian** model adds vega risk. The strong negative correlation between S&P returns and VIX ($\rho \approx -0.75$) means that market drops coincide with volatility spikes, which increase option values and partially offset delta losses.
3. The **copula + Student-t** model preserves the same dependence structure but uses heavier tails. This better captures extreme market movements, generally producing more conservative (higher) VaR and ES estimates.

4. The **volatility surface** model assigns option-specific implied volatilities based on moneyness and maturity, rather than using a flat VIX. This captures the smile/skew effect and is the most realistic model.